Lesson plan · for the teacher

# Evaluate before you trust

AI agents · Level 4 · Lesson 3

**Time**
40 min

**Materials**
Paper and pens for writing test requests; A sheet listing the ten tools and what each needs; A tally table on the board: English and Arabic, trained and new wordings

## Learning goals

- Build a test set with known right answers.
- Score an agent exactly, and break the score down.
- Set a pass mark before you run the test, and test again after changes.

## Activity: The agent's exam

1. Warm-up (5 min): Ask: what makes an exam fair? Collect ideas: questions we know the answers to, the same marking for everyone, some questions not seen in class.
2. Teach (8 min): Explain that an agent is tested the same way. Show the gap: 96–98% on wordings it was trained on, 78–80% on new ones. Explain exact scoring: the right tool and every detail.
3. Write (15 min): Each group writes five requests with the right answer beside each, in both languages, including one new wording, one with a missing detail and one the agent should refuse.
4. Check (8 min): Groups swap pages, check each answer and mark unclear ones. Together, choose the best twenty as the class test set.
5. Wrap-up (4 min): Agree a pass mark for a low-stakes job, and say which jobs need a person's yes whatever the score. Then give the exit ticket.

**Prepare the sheet of the ten tools and what each needs (place and day for the weather, class and day for the timetable…). A phone is optional, for trying a few requests in the labs.**

## Discussion

- Which jobs at our school need a person's yes whatever the score?
- Why does a company's accuracy figure need the question "on what test?"?
- How often should we run our test set again?

## Exit ticket

- Write one test request with its exact right answer, and one pass mark you would set, with the job it is for.

**Try it at school tomorrow**
Let pupils write the agent's exam: each group writes five requests with the right tool and details, in both languages, including one new wording and one request it should refuse. Keep the best twenty as your school's test set, and use them before you trust any agent.

Practice questions · for you

# Evaluate before you trust

AI agents · Level 4 · Lesson 3

**1** **After teaching it a new tool, the new tool works. What must you test before you use it?**

◯ Only the new tool ◯ The new tool and every old skill ◯ Nothing: the training finished

**2** **Put the steps of an evaluation in order.**

☐ Run every request and score the tool and every detail

☐ Collect real requests with known right answers

☐ Add new wordings, both languages and ones to refuse

☐ Run the same test again after every change

☐ Read every failure and decide what to guard

**3** **What does this program print?**

```
right = {"trained": 16,
         "new": 13}
asked = {"trained": 16,
         "new": 16}
for k in right:
    n = asked[k]
    print(k, right[k], "/", n)
```

◯ 29 / 32 ◯ trained 16 / 16, then new 16 / 16 ◯ trained 16 / 16, then new 13 / 16

**4** **Complete the sentence.**

A fair test of an agent uses wordings it was _____ on and _____ wordings, and runs again after every _____.

**Word bank:** new · praised · holiday · trained · change

**5** **An agent that scored 29 of 32 (91%) will be right about 91% of the time on everything staff ask it.**

◯ True ◯ False

✎ **Think of an AI tool your school uses or is offered. What would you put in its test set, and what pass mark would you set before trusting it?**

Check your answers

# Evaluate before you trust

AI agents · Level 4 · Lesson 3

**1** **The new tool and every old skill**

Learning something new can make a model forget: with no old examples mixed in, Darsi Tiny kept only 1 of 8 old skills.

**2** **Top to bottom: 3, 1, 2, 5, 4**

First the requests, then the hard cases, then the same exact scoring for all, then every failure, which tells you what to guard or not use the agent for.

**3** **trained 16 / 16, then new 13 / 16**

The loop prints each kind with its right answers and the requests asked: 16 of 16 on trained wordings but 13 of 16 on new ones.

**4** **A fair test of an agent uses wordings it was trained on and new wordings, and runs again after every change.**

Trained wordings show what it learned; new wordings show how it does with real people; and running the test again after every change catches skills that got worse.

**5** **False**

32 requests is a small test: one more mistake moves the score by about 3 points.